

Chapter 4

Mathematical Foundations

4.1 Introduction

Before introducing the formal structure of the Theory of Entropicity (ToE), it is necessary to establish the mathematical foundations upon which the theory is built. These foundations arise from three major disciplines:

1. Differential Geometry.
2. Statistical Mechanics and Thermodynamics.
3. Information Geometry.

The Theory of Entropicity draws heavily from each of these areas. Differential geometry provides the language of spacetime and gravitation. Thermodynamics provides the conceptual basis for entropy and irreversibility. Information geometry provides the mathematical framework through which informational structures acquire geometric meaning.

The purpose of this chapter is to develop the essential concepts and notation required for the subsequent formulation of the Hybrid Manifold Architecture System (HMAS), the Obidi Convention, and the Local Obidi Action.

Part I

Differential Geometry

4.2 Smooth Manifolds

A manifold is a mathematical structure that locally resembles Euclidean space while potentially possessing a more complicated global structure.

Let

$$M$$

be a smooth four-dimensional manifold.

Each point on the manifold is described by coordinates

$$x^\mu, \quad \mu = 0, 1, 2, 3.$$

The collection

$$(x^0, x^1, x^2, x^3)$$

forms a local coordinate chart.

In relativistic physics these coordinates are typically identified as

$$x^\mu = (ct, x, y, z).$$

The manifold itself contains no notion of distance, angle, or curvature.

Those properties arise only after introducing a metric tensor.

4.3 Metric Tensor

The metric tensor is a symmetric rank-two tensor

$$g_{\mu\nu}.$$

It defines the infinitesimal interval

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu.$$

For Lorentzian spacetime the metric signature is

$$(-, +, +, +).$$

The metric determines:

- distances,
- angles,
- causal structure,
- proper time,
- spacetime volume.

The inverse metric

$$g^{\mu\nu}$$

satisfies

$$g^{\mu\rho} g_{\rho\nu} = \delta^\mu_\nu,$$

where

$$\delta^\mu_\nu$$

is the Kronecker delta.

The metric allows indices to be raised and lowered:

$$V_\mu = g_{\mu\nu} V^\nu,$$

$$V^\mu = g^{\mu\nu} V_\nu.$$

This operation is fundamental throughout gravitational theory.

4.4 Covariant Differentiation

Ordinary derivatives fail to transform correctly under general coordinate transformations.

To preserve tensorial structure one introduces the covariant derivative

$$\nabla_\mu.$$

The connection coefficients are the Christoffel symbols

$$\Gamma_{\mu\nu}^\rho = \frac{1}{2} g^{\rho\sigma} (\partial_\mu g_{\nu\sigma} + \partial_\nu g_{\mu\sigma} - \partial_\sigma g_{\mu\nu}).$$

The covariant derivative of a vector field is

$$\nabla_\mu V^\nu = \partial_\mu V^\nu + \Gamma_{\mu\rho}^\nu V^\rho.$$

The Levi-Civita connection is uniquely characterized by:

$$\nabla_\rho g_{\mu\nu} = 0$$

(metric compatibility)

and

$$\Gamma_{\mu\nu}^\rho = \Gamma_{\nu\mu}^\rho$$

(torsion-free condition).

These properties ensure consistency between geometry and differentiation.

4.5 Curvature

The defining feature of gravitation in General Relativity is curvature.

Curvature arises from the noncommutativity of covariant derivatives.

The Riemann curvature tensor is defined by

$$R^\rho_{\sigma\mu\nu} = \partial_\mu \Gamma^\rho_{\nu\sigma} - \partial_\nu \Gamma^\rho_{\mu\sigma} + \Gamma^\rho_{\mu\lambda} \Gamma^\lambda_{\nu\sigma} - \Gamma^\rho_{\nu\lambda} \Gamma^\lambda_{\mu\sigma}.$$

The Riemann tensor measures how vectors change when parallel transported around infinitesimal loops.

It contains complete information about local curvature.

4.6 Ricci Tensor

Contracting the Riemann tensor yields the Ricci tensor:

$$R_{\mu\nu} = R^\rho_{\mu\rho\nu}.$$

The Ricci tensor measures volume distortion induced by curvature.

Regions with positive Ricci curvature tend to focus neighboring geodesics, whereas negative Ricci curvature tends to separate them.

4.7 Ricci Scalar

A further contraction yields the Ricci scalar

$$R = g^{\mu\nu} R_{\mu\nu}.$$

The Ricci scalar provides a single scalar measure of spacetime curvature.

It appears prominently in gravitational action principles.

4.8 Einstein Tensor

The Einstein tensor is defined by

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R.$$

The Einstein tensor possesses a crucial property:

$$\nabla^\mu G_{\mu\nu} = 0.$$

This identity guarantees local conservation of energy and momentum.

The Einstein tensor therefore provides the natural geometric object appearing in gravitational field equations.

4.9 Einstein Field Equations

The field equations of General Relativity are

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}.$$

These equations describe the relationship between geometry and matter.

The terms have the following meanings:

$$G_{\mu\nu}$$

= spacetime curvature,

$$\Lambda$$

= cosmological constant,

$$T_{\mu\nu}$$

= stress-energy tensor,

$$G$$

= Newton's gravitational constant.

The Theory of Entropicity must reproduce these equations in an appropriate limit.

Their recovery will serve as one of the principal consistency tests of the theory.

Part II

Statistical Mechanics and Thermodynamics

4.10 Entropy

Entropy is one of the most important quantities in physics.

For a discrete probability distribution

$$\{p_i\},$$

the Gibbs-Shannon entropy is

$$S = -k_B \sum_i p_i \ln p_i.$$

For a continuous distribution

$$p(x),$$

the entropy becomes

$$S = -k_B \int p(x) \ln p(x) dx.$$

Entropy quantifies uncertainty, multiplicity, and information content.

Within conventional physics it is usually regarded as a derived quantity.

The Theory of Entropicity seeks to elevate entropy to a more fundamental status.

4.11 The Second Law

The Second Law of Thermodynamics states

$$\Delta S \geq 0.$$

Entropy tends to increase in isolated systems.

This law introduces an arrow of time into physical processes.

The existence of temporal asymmetry despite fundamentally time-symmetric microscopic laws is one of the central motivations for entropic approaches to physics.

4.12 Clausius Relation

The infinitesimal relation between heat and entropy is

$$\delta Q = T dS.$$

This equation forms the basis of thermodynamic reasoning.

It played a pivotal role in Jacobson's derivation of Einstein's equations.

In the context of ToE, it serves as an important conceptual bridge between thermodynamics and gravitation.

4.13 Black Hole Entropy

The entropy of a black hole is

$$S_{BH} = \frac{k_B A}{4\ell_P^2}.$$

Since

$$\ell_P^2 = \frac{G\hbar}{c^3},$$

one obtains

$$S_{BH} = \frac{k_B c^3 A}{4G\hbar}.$$

This remarkable equation links:

- gravity,
- quantum theory,
- thermodynamics,
- geometry.

It constitutes one of the strongest motivations for the Theory of Entropicity.

Part III

Information Geometry

4.14 Information as Geometry

Information geometry studies probability distributions as geometric objects.

Instead of treating a probability distribution merely as a collection of numbers, one views it as a point on a manifold.

Distances on this manifold quantify distinguishability between statistical states.

This insight provides one of the foundational pillars of ToE.

4.15 Relative Entropy

The Kullback–Leibler divergence between distributions $p(x)$ and $q(x)$ is

$$D_{KL}(p||q) = \int p(x) \ln \left(\frac{p(x)}{q(x)} \right) dx.$$

Although not a true metric, the KL divergence measures informational distinguishability.

Many information-geometric structures emerge from its local expansion.

4.16 Fisher Information

Consider a parametric family

$$p(x|\theta).$$

The Fisher information matrix is

$$I_{ij} = \int p(x|\theta) \left(\frac{\partial \ln p}{\partial \theta^i} \right) \left(\frac{\partial \ln p}{\partial \theta^j} \right) dx.$$

The Fisher information measures the sensitivity of a distribution to parameter changes.

4.17 Fisher–Rao Metric

The Fisher–Rao metric is

$$g_{ij} = I_{ij}.$$

Explicitly,

$$g_{ij} = \int p(x|\theta) \frac{\partial \ln p}{\partial \theta^i} \frac{\partial \ln p}{\partial \theta^j} dx.$$

This metric transforms the statistical parameter space into a Riemannian manifold.

The Fisher–Rao sector forms the first informational sector of the Hybrid Manifold Architecture System.

4.18 Fubini–Study Metric

Quantum mechanics possesses an analogous geometry.

For normalized quantum states

$$|\psi\rangle,$$

the Fubini–Study line element is

$$ds_{FS}^2 = 4 \left(\langle d\psi | d\psi \rangle - |\langle \psi | d\psi \rangle|^2 \right).$$

This metric quantifies distinguishability between neighboring quantum states.

The Fubini–Study sector forms the second informational sector of HMAS.

4.19 Information Geometry and Gravity

The significance of information geometry lies in a simple observation.

Information possesses geometry.

General Relativity identifies gravity with geometry.

Therefore,

$$\text{Information} \longrightarrow \text{Geometry} \longrightarrow \text{Gravity}.$$

This conceptual chain serves as one of the principal motivations for the Theory of Entropicity.

4.20 Transition to the Hybrid Manifold

The preceding developments reveal three distinct but interconnected geometrical structures:

Fisher–Rao Geometry,

Fubini–Study Geometry,

Lorentzian Geometry.

The next chapter introduces the Hybrid Manifold Architecture System (HMAS), which unifies these three geometries into a single mathematical framework and provides the foundation for the Obidi Convention and the Theory of Entropicity.